

From Mesh Avoidance to Catalan Inflation

A proof of the generating-function conjecture for OEIS A289587

A self-contained proof and reproducibility note

September 20, 2026

Abstract

OEIS A289587 counts permutations avoiding the classical pattern 321 and one of two inverse mesh patterns of length two. Its entry attributes an algebraic generating-function conjecture to Thomas Scheuerle, dated December 23, 2025. We prove that formula by identifying exactly what a mesh occurrence means inside a 321-avoiding permutation. Every occurrence is either a pair of fixed points or a pair of entries in an increasing run of consecutive excedance values. Avoidance therefore means at most one fixed point and no upper succession. Contracting maximal upper-succession runs is a reversible operation; at the level of the Narayana generating function it becomes the substitution $u \mapsto u/(1+xu)$. A direct-sum decomposition then gives the conjectured radical expression. We also derive a full mesh-occurrence statistic, multivariate refinements, positive coefficient recurrences and finite sums, an algorithm computing the first N terms with $O(N)$ integer arithmetic operations, and the precise asymptotic $a_n \sim 0.4139310680 \dots (3.5464554447 \dots)^n n^{-3/2}$ together with its $1/n$ correction. The proof is self-contained, and the two accompanying independent exact-arithmetic programs check the literal mesh definition independently of the generating function.

Keywords. Permutation patterns; mesh patterns; Catalan and Narayana numbers; excedances; fixed points; algebraic generating functions.

1 The conjecture and the result

The classical pattern 321 occurs in a permutation $\pi = \pi_1 \cdots \pi_n$ when there are indices $i < j < k$ with $\pi_i > \pi_j > \pi_k$. Write $\text{Av}_n(321)$ for the permutations with no such triple. This is a central Catalan family; the excedance structure and its connection with lattice paths have been studied in detail, for example by Reifegerste [5]. We will prove the particular facts needed here rather than assume them.

A mesh pattern adds empty-region conditions to an ordinary pattern. For the underlying pattern 12, select $i < j$ with $a = \pi_i < b = \pi_j$. The vertical lines through i, j and horizontal lines through a, b divide all other permutation points into nine open rectangles. Label their columns and rows 0, 1, 2, from left to right and from bottom to top. A pair is an occurrence of $(12, R)$ if every rectangle indexed by $(c, r) \in R$ is empty. Points on the selected lines cause no ambiguity, since a permutation has distinct positions and distinct values. This is the usual mesh-pattern framework of Brändén and Claesson [3], specialized to a length-two pattern.

The shading sets relevant to A289587 are

$$R_{174} = \{(0, 1), (0, 2), (1, 0), (1, 2), (2, 1)\}, \quad (1.1)$$

$$R_{234} = \{(0, 1), (1, 0), (1, 2), (2, 0), (2, 1)\}. \quad (1.2)$$

These coordinates agree with the encoding described in the linked SeqFan thread [2], which assigns the cell (c, r) the weight 2^{3c+r} ; summing the weights of the shaded cells gives the displayed labels 174 and 234. Those labels only name the patterns and play no arithmetic role in the proof. The explicit sets also remove any dependence of our proof or programs on an unexplained numerical pattern identifier.

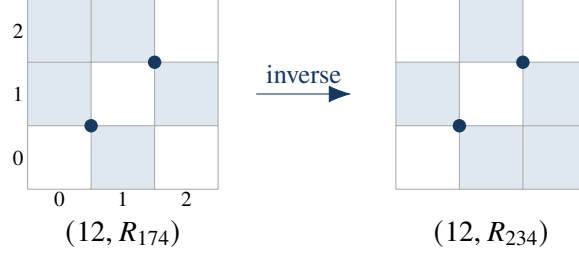


Figure 1: Shaded rectangles must contain no other permutation points. Transposition of the diagram interchanges the two mesh patterns.

Let a_n count $\pi \in \text{Av}_n(321)$ avoiding $(12, R_{174})$, including the empty permutation at $n = 0$.

Lemma 1.1 (Inverse equivalence). *The map $\pi \mapsto \pi^{-1}$ is a bijection between the class of 321-avoiders avoiding $(12, R_{174})$ and the class of 321-avoiders avoiding $(12, R_{234})$.*

Proof. Inverting a permutation transposes its point plot. The pattern 12 is invariant under transposition, the classical pattern 321 is its own inverse, and transposing every cell of R_{174} gives exactly R_{234} . Thus inversion preserves 321-avoidance and exchanges the two mesh restrictions, in both directions. $\square$

Both versions in the OEIS definition therefore have the same counting sequence. This equivalence is proved, not inferred from agreement of initial terms. Refinements by excedances below refer to the $(12, R_{174})$ class; the inverse bijection need not preserve that statistic.

The OEIS entry [1] lists

$$1, 1, 1, 3, 6, 18, 47, 139, 405, 1225, 3740, 11602, \\ 36357, 115049, 366969, 1178791, 3809802.$$

for $n = 0, \dots, 16$, and marks the following formula as conjectural.

Theorem 1.2 (The A289587 generating function). *Set*

$$D(x) = x^4 - 2x^3 - 5x^2 - 2x + 1, \\ P(x) = x^4 + 4x^3 + 11x^2 + 10x + 3, \quad Q(x) = x^2 + 5x + 3.$$

Then

$$A(x) := \sum_{n \geq 0} a_n x^n = \frac{P(x) - Q(x)\sqrt{D(x)}}{8x(1+x)^2}. \quad (1.3)$$

Here $\sqrt{D(x)}$ is the formal square root with constant term 1; the apparent singularity at $x = 0$ is removable.

The interesting issue is not recognizing a square root from these numbers. A mesh condition quantifies over two selected entries and the emptiness of several surrounding rectangles. It can detect arbitrarily distant entries, and so need not reduce to a local prohibition. Here it does not: besides a local obstruction there is a genuinely global one, namely two singleton direct-sum components, even when they are separated by a nontrivial component. The proof below isolates that obstruction before any generating-function manipulation is performed.

Status and scope. The entry, consulted on September 20, 2026, still labels (1.3) “Conjectured g.f.” and credits Scheuerle on December 23, 2025. The proof below establishes the exact identity, not merely an initial coefficient agreement. I did not locate a proof of this formula in the entry or its linked discussion. This is a proof of the OEIS-listed conjecture; we make no claim that an exhaustive literature search has ruled out every independent prior derivation. Standard Catalan and Narayana ingredients are explicitly separated from the mesh-specific argument.

2 Elementary structure of 321-avoiders

An index i is an *excedance* of π if $\pi_i > i$, and a fixed point if $\pi_i = i$. Let $\text{exc}(\pi)$ and $\text{fix}(\pi)$ denote their numbers. An entry is a left-to-right maximum if it exceeds every preceding entry.

Lemma 2.1 (Two increasing subsequences). *In a 321-avoiding permutation, the entries at excedance positions form an increasing subsequence, and the entries at nonexcedance positions form an increasing subsequence. Conversely, these two increasing-subsequence conditions imply 321-avoidance. Every excedance entry is a left-to-right maximum.*

Proof. Suppose $i < j$ are excedances with $\pi_i > \pi_j$. Since $\pi_j > j$, not all values smaller than π_j can occupy the first $j - 1$ positions. Some $k > j$ has $\pi_k < \pi_j$, giving a forbidden triple. The same argument proves the last assertion: if j is an excedance and an earlier entry is larger than π_j , a smaller later value completes a 321.

Now suppose $i < j$ are nonexcedances with $\pi_i > \pi_j$. There are only $\pi_i - 1$ values below π_i , and at least one of them, namely π_j , is later than position i . Because $i \geq \pi_i$, the $i - 1$ preceding positions cannot all contain values below π_i . Some $h < i$ therefore has $\pi_h > \pi_i > \pi_j$, again impossible.

For the converse, any three entries include two belonging to the same one of the two indicated subsequences. Those two entries are increasing, so the three entries cannot be decreasing. $\square$

Lemma 2.2 (A fixed point separates the permutation). *If $\pi \in \text{Av}_n(321)$ and $\pi_i = i$, then all earlier values are less than i and all later values are greater than i .*

Proof. If an earlier value exceeded i , counting the values $1, \dots, i - 1$ would force a later value below i . Together with $\pi_i = i$ these would form a 321. Thus all $i - 1$ preceding values are below i ; they are exactly $1, \dots, i - 1$, and every later value is larger. $\square$

For permutations α of length r and β of length s , their *direct sum* is

$$\alpha \oplus \beta = \alpha_1 \cdots \alpha_r (\beta_1 + r) \cdots (\beta_s + r).$$

A nonempty permutation is sum-indecomposable if it is not a direct sum of two nonempty permutations. Every permutation has a unique decomposition into sum-indecomposable blocks: cut precisely after those $j < n$ for which the first j values form the set $\{1, \dots, j\}$.

By theorem 2.2, fixed points of a 321-avoider are exactly the singleton blocks in this decomposition. No triple 321 straddles a direct-sum boundary, so each block is itself a 321-avoider. These facts will make the fixed-point restriction particularly simple. We use “fixed point” and “singleton direct-sum component” interchangeably from now on; inside $\text{Av}(321)$ they are the same thing.

The class can also be described using left-to-right maxima instead of excedances, and the literature on 321-avoiders uses both. The two descriptions are not interchangeable as local conditions, and the next lemma records exactly where they agree. It is used in theorem 3.4 and in the second verification program, and nowhere in the main line of the proof.

Lemma 2.3 (Records, excedances and fixed points). *Let $\pi \in \text{Av}_n(321)$ and write $\text{rec}(\pi)$ for its number of left-to-right maxima. Then:*

- (i) π_i is a left-to-right maximum if and only if i is an excedance or a fixed point. Hence $\text{rec}(\pi) = \text{exc}(\pi) + \text{fix}(\pi)$.
- (ii) A sum-indecomposable π of length at least two has no fixed point. On such a permutation the left-to-right maxima are exactly the excedance entries, so $\text{rec}(\pi) = \text{exc}(\pi)$.
- (iii) If π_i is a left-to-right maximum, $\pi_{i+1} = \pi_i + 1$, and there is a sum cut at i , then the entries at positions i and $i + 1$ are two adjacent singleton components.
- (iv) Within a direct-sum component, being a left-to-right maximum is equivalent to being one after standardizing that component.

Proof. (i) An excedance entry is a left-to-right maximum by theorem 2.1; a fixed point is one by theorem 2.2. Conversely, if π_i exceeds all $i - 1$ preceding entries then $\pi_i \geq i$, so i is an excedance or a fixed point. The two cases are disjoint.

(ii) A fixed point is a singleton block, so an indecomposable permutation of length at least two has none, and (i) applies.

(iii) A sum cut at i means $\{\pi_1, \dots, \pi_i\} = \{1, \dots, i\}$, so the largest of the first i entries is i ; since π_i is a left-to-right maximum, $\pi_i = i$ and therefore $\pi_{i+1} = i + 1$. Then $\{\pi_1, \dots, \pi_{i-1}\} = \{1, \dots, i - 1\}$ and $\{\pi_1, \dots, \pi_{i+1}\} = \{1, \dots, i + 1\}$, so $i - 1$ and $i + 1$ are sum cuts as well and both entries are singleton components.

(iv) Every earlier component consists of values smaller than every value of the current one, so it neither creates nor destroys a maximum. $\square$

3 The exact mesh-occurrence statistic

Definition 3.1. An *upper succession*, also called an upper bond in this note, is a pair of adjacent positions $i, i + 1$ satisfying

$$\pi_i > i, \quad \pi_{i+1} = \pi_i + 1. \quad (3.1)$$

Both positions are then excedances. A maximal upper run is a maximal consecutive block of excedance entries whose values rise by one at each step. Excedance entries not belonging to a longer run count as runs of length one.

Theorem 3.2 (Occurrence formula). *Let $\pi \in \text{Av}_n(321)$ have f fixed points and maximal upper-run lengths $\ell_1, \dots, \ell_k$. Then*

$$\text{occ}_{(12, R_{174})}(\pi) = \binom{f}{2} + \sum_{j=1}^k \binom{\ell_j}{2}. \quad (3.2)$$

More precisely, its mesh occurrences are exactly the pairs of fixed points and the pairs of entries lying in the same maximal upper run.

Proof. For an occurrence with endpoints (i, a) and (j, b) , the shading (1.1) says exactly that

$$\pi_h < a \quad \text{for every } h < i, \quad (3.3)$$

$$a < \pi_h < b \quad \text{for every } i < h < j, \quad (3.4)$$

$$\pi_h \notin (a, b) \quad \text{for every } h > j. \quad (3.5)$$

Every value strictly between a and b must therefore occur strictly between positions i and j , and every entry in those positions has such a value. Consequently

$$j - i = b - a, \quad \{\pi_i, \dots, \pi_j\} = \{a, a + 1, \dots, b\}. \quad (3.6)$$

Also (3.3) forces $a \geq i$.

If $a = i$, then (3.6) gives $b = j$: the endpoints are fixed points. Conversely, any pair of fixed points satisfies (3.3)–(3.5) by theorem 2.2.

If $a > i$, at least one value $c < a$ is missing from the $i - 1$ positions before i . It cannot lie between i and j by (3.4); hence it lies after j . An inversion anywhere in the block from i to j , followed by this value c , would create a 321. The whole block is therefore increasing, and (3.6) makes it

$$a, a + 1, \dots, b.$$

Every one of these positions is an excedance because the value minus the position is the constant $a - i > 0$. Thus the endpoints lie in one upper run.

Conversely, take two entries in one upper run. The first is a left-to-right maximum by theorem 2.1. All intermediate values occur consecutively between the endpoints, so none can occur later. Thus (3.3)–(3.5) hold. The two types of occurrence are disjoint, and counting endpoint pairs proves (3.2). $\square$

Corollary 3.3 (The mesh condition without a mesh diagram). *A permutation $\pi \in \text{Av}_n(321)$ avoids $(12, R_{174})$ if and only if it has no upper succession and has at most one fixed point.*

This is the decisive step: a condition involving five empty rectangles has become one local prohibition and one global restriction on singleton sum blocks. Equivalently, and this is the form in which the criterion is most often stated, a 321-avoider *contains* $(12, R_{174})$ exactly when it has an upper succession or at least two singleton direct-sum components. Theorem 3.2 is strictly stronger: theorem 3.3 is its specialization to $\text{occ} = 0$.

Remark 3.4 (Two local conditions that are not the same condition). A variant of (3.1) appears naturally if one marks left-to-right maxima instead of excedances: call i a *record bond* when π_i is a left-to-right maximum and $\pi_{i+1} = \pi_i + 1$. These are genuinely different predicates. The permutation 123 has the record bonds 1, 2 and 2, 3 but no upper succession at all, since every entry is a fixed point; among the 321-avoiders of length at most six there are 31 on which the two predicates disagree. What is true is the following. By theorem 2.3(i), a record bond that is not an upper succession starts at a fixed point $\pi_i = i$ and then $\pi_{i+1} = i + 1$, which by theorem 2.3(iii) makes both entries singleton components; so a record bond that is not an upper succession forces a second fixed point. Hence

$$\text{no upper succession and } \text{fix} \leq 1 \iff \text{no record bond and } \text{fix} \leq 1,$$

and the two conjunctions cut out the same class, even though neither local condition implies the other. On the nontrivial sum-indecomposable 321-avoiders, where the contraction of section 4 is actually performed, theorem 2.3(ii) makes the two predicates literally identical, because there are no fixed points there. This is why the record-refined and excedance-refined core series of section 7 are the same series and not a shift of one another; both descriptions are correct, and the merged notation below uses excedances throughout.

Example 3.5. The five 321-avoiders of length three are 123, 132, 213, 231, 312. The first has three fixed points, while 231 has the upper succession 2, 3. The remaining three are accepted, giving $a_3 = 3$. The permutation $1324 = 1 \oplus 21 \oplus 1$ has no upper succession, but its two fixed points, 1 and 4, form a mesh

occurrence. Thus the fixed-point condition cannot be omitted; note that the two singleton blocks producing it are not adjacent, so banning adjacent singletons would not suffice either.

By contrast, 4123 is accepted. Its consecutive ascents 1, 2 and 2, 3 are neither excedance successions nor ascents beginning at a left-to-right maximum, and it is sum-indecomposable with no fixed point. This is the example to keep in mind: plain succession-avoidance is the wrong local condition, and the word “upper” in (3.1) is doing real work.

4 Contracting upper runs

The next step is an exact decomposition, not an inclusion–exclusion approximation. Contract every maximal upper run to one entry and standardize the resulting word, meaning that its smallest remaining value becomes 1, the next becomes 2, and so on. Call the result the *upper-run core*.

Theorem 4.1 (Canonical core and inflation). *Every 321-avoiding permutation has a unique representation as a 321-avoiding core with no upper succession, with each excedance entry of the core inflated into a nonempty increasing interval. The lengths of the inflated intervals are precisely the original maximal upper-run lengths. This operation preserves the number of fixed points and preserves the direct-sum block structure.*

Proof. We first justify a single contraction. Suppose adjacent positions $i, i + 1$ contain $a, a + 1$ with $a > i$. Delete the second entry and standardize. Pattern avoidance survives taking a subpermutation. All entries before i are below a by theorem 2.1, so they are unchanged. The surviving entry a remains an excedance.

For an entry later than $i + 1$, if its value is larger than $a + 1$, both its position and value decrease by one; its excedance or fixed-point status is unchanged. Suppose instead that this later entry is $b < a$ at position j . Every value smaller than b must precede j : otherwise a, b , and a later smaller value would form a 321. The $j - 1$ preceding positions contain those $b - 1$ smaller values and also both $a, a + 1$. Therefore $j - 1 \geq b + 1$, or $b \leq j - 2$. After contraction, $b < j - 1$, so it remains a strict nonexcedance. Thus precisely one excedance disappears, and no fixed point appears or disappears.

It remains important to check that contraction creates no unexpected new upper bond. If two consecutive excedance entries have a gap between their values, every value in that gap occurs later: the first entry is a left-to-right maximum and there is no position between the two. Such intervening values cannot be later excedances, since excedance values increase; they are nonexcedances and are never contracted. Their presence therefore prevents the gap from disappearing. Likewise, a nonexcedance between two upper runs is never deleted, so separated runs cannot become adjacent. Contracting a bond within a run merely shortens that run. Hence the maximal runs have a uniquely determined bond-free core, independent of contraction order.

Conversely, inflate any excedance of a bond-free core into a consecutive increasing block. Increasing interval inflation preserves 321-avoidance: a decreasing triple uses at most one entry from such a block, and would project to a decreasing triple in the core. Entries before the inflated point have smaller values and are unchanged; later larger entries have both position and value shifted by the same amount, while later smaller entries stay strict nonexcedances. Fixed points remain fixed, and all entries of the inflated block are excedances.

A pre-existing position gap between core excedances contains a retained nonexcedance; a pre-existing value gap contains a retained nonexcedance value, by the preceding argument. Hence different inflated blocks never merge into one upper run. The inflation lengths and the core are therefore recovered uniquely by contraction.

Finally, an upper bond cannot cross a direct-sum boundary. If it did, its first value, being a left-to-right maximum and lying in the left summand, would have to equal the boundary position, contradicting strict excedance. Inflating an entry of a nontrivial indecomposable block preserves indecomposability: a sum

cut between distinct inflated blocks would project to a sum cut in the core; a cut inside an inflated block is impossible because its first entry still exceeds its position and leaves a smaller value on the right. Thus the sum blocks correspond under contraction and inflation. $\square$

Example 4.2. The permutation

$$2341675 = 2341 \oplus 231$$

has upper runs of lengths 3 and 2. Its core is $2143 = 21 \oplus 21$. Inflating the two excedances of 2143 by those lengths reconstructs 2341675. The occurrence formula gives $\binom{3}{2} + \binom{2}{2} = 4$ mesh occurrences.

5 The Narayana generating function, from first principles

Define

$$C(x, u) = \sum_{n \geq 0} \sum_{\pi \in \text{Av}_n(321)} x^n u^{\text{exc}(\pi)}. \quad (5.1)$$

We need its functional equation with u still present; substituting the ordinary Catalan series $C(x, 1)$ too early would lose exactly the information needed for upper-run contraction.

Lemma 5.1 (Excedance arrays and Dyck valleys). *Permutations in $\text{Av}_n(321)$ with k excedances are in bijection with Dyck paths of semilength n having k valleys.*

Proof. Let $e_1 < \dots < e_k$ be the excedance positions and $v_1 < \dots < v_k$ their values. By theorem 2.1, the remaining values occupy the remaining positions in increasing order. The pair of arrays therefore determines the permutation, and satisfies $e_j < v_j$.

Conversely, choose increasing arrays in $\{1, \dots, n\}$ with $e_j < v_j$, put v_j in position e_j , and fill the remaining positions increasingly. For every t , the number of selected positions at most t is at least the number of selected values at most t , since $v_j \leq t$ implies $e_j < t$. Thus the number of unselected values at most t is at least the number of unselected positions at most t . Increasing matching places each remaining value at a position at least as large as itself. These are precisely the nonexcedances, and the two increasing subsequences imply 321-avoidance.

Use north and east steps for a Dyck path from $(0, 0)$ to (n, n) staying in $y \geq x$. Specify its valleys (an east step followed by a north step) to be

$$(e_1, v_1 - 1), \dots, (e_k, v_k - 1).$$

Both coordinates increase strictly, and $e_j \leq v_j - 1$. These valleys uniquely determine a path: alternate maximal north and east runs, starting with a north run and ending with an east run. Each valley is on or above the diagonal, so the whole path stays above it. Conversely, the valleys of such a path recover exactly the allowed arrays. The path with no valleys is $N^n E^n$, corresponding to the identity permutation. $\square$

Proposition 5.2 (Narayana equation). *The series $C(x, u)$ is the unique formal solution with constant term 1 of*

$$C = 1 + xC(1 - u + uC). \quad (5.2)$$

Equivalently, for $n \geq 1$,

$$[x^n][u^k]C(x, u) = \frac{1}{n} \binom{n}{k} \binom{n}{k+1}, \quad 0 \leq k < n. \quad (5.3)$$

Proof. A nonempty Dyck path decomposes uniquely as $NPEQ$, where P, Q are Dyck paths. There is a new valley between the indicated E and the start of Q exactly when Q is nonempty. Thus the generating function is $1 + xC(1 + u(C - 1))$, proving (5.2). The factor x determines each coefficient from earlier ones, giving uniqueness.

For the coefficient formula put $Y = C - 1$. Then $Y = x(1 + Y)(1 + uY)$. Formal Lagrange inversion [4] gives

$$[x^n]Y = \frac{1}{n} [z^{n-1}] (1 + z)^n (1 + uz)^n.$$

Extracting u^k proves (5.3). This recovers the classical Narayana distribution of excedances (cf. [5, Corollary 3.7]) with the indexing conventions used here. $\square$

6 From contraction to the conjectured radical

Let $B(x, u)$ count all upper-bond-free 321-avoiders by length and excedance number, with any number of fixed points. A core excedance has weight xu . Inflating it into a positive run changes this weight to

$$xu + (xu)^2 + (xu)^3 + \cdots = \frac{xu}{1 - xu}.$$

The canonical decomposition in theorem 4.1 gives

$$C(x, u) = B\left(x, \frac{u}{1 - xu}\right), \quad \boxed{B(x, u) = C\left(x, \frac{u}{1 + xu}\right)}. \quad (6.1)$$

The two substitutions are compositional inverses as formal power series in x . No analytic convergence assertion is involved: every coefficient of x on either side is a polynomial in u , and the substituted denominators have constant term 1. The alternating signs produced by $u \mapsto u/(1 + xu)$ express the inversion of a bijection that has been proved, not an unmotivated guess and not an assumed inclusion–exclusion rule.

At $u = 1$, write simply $B(x) = B(x, 1)$. Substitution into (5.2) yields

$$xB^2 - (1 + x - x^2)B + 1 + x = 0, \quad B(x) = \frac{1 + x - x^2 - \sqrt{D(x)}}{2x}. \quad (6.2)$$

The other algebraic branch has a pole at zero, whereas this branch has constant term 1.

Let $I(x)$ count the nontrivial sum-indecomposable blocks in this bond-free class. Upper bonds do not cross sum boundaries, so arbitrary bond-free permutations are sequences of singleton blocks, of weight x , and nontrivial blocks, counted by I . Therefore

$$B = \frac{1}{1 - x - I}, \quad I = 1 - x - \frac{1}{B} = \frac{1 - x - x^2 - \sqrt{D(x)}}{2(1 + x)}. \quad (6.3)$$

The empty permutation is included in B , but not in I .

Let $H(x)$ count the bond-free permutations with no fixed points. They are sequences of nontrivial blocks, so

$$H = \frac{1}{1 - I} = \frac{B}{1 + xB} = \frac{1 + 3x + x^2 - \sqrt{D(x)}}{4x(1 + x)}. \quad (6.4)$$

By theorem 3.3, an accepted permutation either has no fixed point, or has one fixed point with an arbitrary fixed-point-free bond-free permutation on either side. It follows that

$$\boxed{A(x) = H(x) + xH(x)^2}. \quad (6.5)$$

This identity has a literal bijective interpretation: split at the unique singleton block when one exists. The term xH^2 counts an actual singleton direct-sum component, not a marked arbitrary position; written in terms of I through $H = 1/(1 - I)$, it says that a permutation with m nontrivial blocks offers exactly $m + 1$ places for that singleton.

Completion of the proof of theorem 1.2. Substitute (6.4) into (6.5) and replace $(\sqrt{D})^2$ by D . The rational part of the resulting numerator is $P(x)$ and the coefficient of $\sqrt{D}$ is $-Q(x)$, with denominator $8x(1+x)^2$, exactly as in (1.3). Every prior identity was derived from a bijection or a unique formal functional equation, so this proves the formula for all coefficients simultaneously. $\square$

Two polynomial certificates for the last step. The final substitution can be checked by hand, without trusting any symbolic simplification. Let r be an indeterminate and put

$$G(x, r) = (1+x)r^2 + (x+x^2-1)r + x^2, \quad \sigma(x, r) = 1 - x - x^2 - 2(1+x)r. \quad (6.6)$$

By (8.1) below, equivalently by (6.3), the series $I(x)$ is the unique formal solution of $G(x, I) = 0$ with $I(0) = 0$. Direct expansion of polynomials gives the two identities

$$\sigma^2 - D(x) = 4(1+x)G(x, r), \quad (6.7)$$

$$(P - Q\sigma)(1-r)^2 - 8x(1+x)^2(1-r+x) = 2(Q(x)r - 3x^2 - 7x - 3)G(x, r). \quad (6.8)$$

Set $r = I(x)$. The right-hand sides vanish. Then (6.7) says $\sigma(x, I)^2 = D(x)$, and $\sigma(x, I)$ has constant term 1, so $\sigma(x, I) = \sqrt{D(x)}$. Since $A = H + xH^2 = (1 - I + x)/(1 - I)^2$ by (6.5) and $H = 1/(1 - I)$, identity (6.8) is exactly $8x(1+x)^2A = P - Q\sqrt{D}$. Both certificates are polynomial identities in two indeterminates; verifying them requires only expansion.

Why the quartic occurs. The discriminant of the Narayana equation becomes

$$(1 + x - x^2)^2 - 4x(1+x) = D(x)$$

after the upper-run substitution. Thus the unusual quartic is not an accident of guessing from a short initial segment: it is the ordinary Catalan discriminant transformed by a canonical run contraction.

7 Refinements beyond avoidance

The same proof retains substantially more information than a single univariate generating function. Define

$$H(x, u) = \frac{B(x, u)}{1 + xB(x, u)}, \quad B(x, u) = C\left(x, \frac{u}{1 + xu}\right). \quad (7.1)$$

This counts the upper-bond-free 321-avoiders with no fixed points, by length and excedances.

Theorem 7.1 (Fixed-point refinement). *The generating function for upper-bond-free 321-avoiders, with t marking fixed points, is*

$$\mathcal{B}(x, u, t) = \frac{H(x, u)}{1 - txH(x, u)}. \quad (7.2)$$

In particular, the subfamily with exactly r fixed points has generating function

$$x^r H(x, u)^{r+1}. \quad (7.3)$$

The excedance-refined generating function for A289587 is $H(x, u) + xH(x, u)^2$.

Proof. Split a permutation at its r singleton sum blocks. This gives an ordered $(r + 1)$ -tuple of fixed-point-free pieces, including empty pieces, and r singleton blocks. The decomposition is unique, and excedance number is additive under direct sum. Summing over r gives (7.2). $\square$

One can also enumerate the number of mesh occurrences itself. Let

$$W(x, u, q) = \sum_{\ell \geq 1} x^{\ell-1} u^\ell q^{\binom{\ell}{2}} = u + xu^2q + x^2u^3q^3 + x^3u^4q^6 + \cdots. \quad (7.4)$$

Multiplying W by x gives the weight of one inflated upper run.

Theorem 7.2 (All mesh occurrences). *For all 321-avoiding permutations, the joint generating function of length, excedances, fixed points, and occurrences of $(12, R_{174})$ is*

$$\begin{aligned} \mathcal{F}(x, u, t, q) &:= \sum_{n \geq 0} \sum_{\pi \in \text{Av}_n(321)} x^n u^{\text{exc}(\pi)} t^{\text{fix}(\pi)} q^{\text{occ}_{(12, R_{174})}(\pi)} \\ &= \sum_{r \geq 0} (tx)^r q^{\binom{r}{2}} H(x, W(x, u, q))^{r+1}. \end{aligned} \quad (7.5)$$

Proof. In a fixed-point-free core, replace the weight xu of each excedance by $xW(x, u, q)$. The inflated run contributes $\binom{\ell}{2}$ occurrences by theorem 3.2. Insert r fixed points as in theorem 7.1; together they contribute $\binom{\ell}{2}$ occurrences, independently of all upper runs. No other occurrences exist. This gives (7.5). $\square$

Every coefficient of x in (7.5) is a polynomial in the other variables, so all substitutions are formally well-defined. At $q = 0$, only run length one and $r = 0, 1$ survive; this recovers the avoidance formula. At $q = 1$, $W = u/(1 - xu)$, undoing core contraction, and at $t = 1$ the result reduces to $C(x, u)$.

Corollary 7.3 (Counting upper bonds rather than mesh occurrences). *If s marks the number of adjacent upper bonds, then the joint generating function for all 321-avoiders is*

$$\frac{C\left(x, \frac{u}{1 + (1-s)xu}\right)}{1 + (1-t)x C\left(x, \frac{u}{1 + (1-s)xu}\right)}. \quad (7.6)$$

Proof. A run of length ℓ has $\ell - 1$ adjacent upper bonds, so its series is $xu/(1 - sxu)$. Apply (6.1) and then (7.2). $\square$

The difference between (7.5) and (7.6) is essential: a length-three run gives two adjacent bonds but three mesh occurrences.

The refined core series

It is worth isolating the excedance refinement of the core series I of (6.3), since the bond-free cores are the objects everything else is assembled from. Set

$$I(x, u) = 1 - x - \frac{1}{B(x, u)}, \quad \text{so that} \quad H(x, u) = \frac{1}{1 - I(x, u)}, \quad (7.7)$$

the generating function of the nontrivial sum-indecomposable 321-avoiders with no upper succession, by length and excedances. That (7.7) is what it claims to be is the refinement of (6.3): a bond-free permutation

is a sequence of singleton blocks, of weight x and no excedance, and nontrivial bond-free indecomposable blocks, and excedance number is additive under direct sum. By theorem 2.3(ii) these cores have no fixed point, so on this class the excedance refinement and the left-to-right-maximum refinement are the same refinement, entry by entry.

Proposition 7.4 (The bivariate core equation). *$I(x, u)$ is the unique formal power series with zero constant term satisfying*

$$(1 + xu) I(x, u)^2 + (x + x^2u - 1) I(x, u) + x^2u = 0. \quad (7.8)$$

At $u = 1$ this is (8.1).

Proof. Let $J(x, u)$ count *all* nontrivial sum-indecomposable 321-avoiders by length and excedances. Blocks are independent, a singleton block has weight x and no excedance, so $C = 1/(1 - x - J)$, that is, $J = 1 - x - 1/C$. Using (5.2) this simplifies to $J = xu(C - 1)$, and substituting $C = 1 + J/(xu)$ back into (5.2) gives

$$J^2 + (x(1 + u) - 1)J + x^2u = 0. \quad (7.9)$$

Now $B(x, u) = C(x, u/(1 + xu))$ by (6.1), so $I(x, u) = 1 - x - 1/B(x, u) = J(x, u/(1 + xu))$ directly from the two definitions. Making that substitution in (7.9) and multiplying by the invertible series $1 + xu$ gives (7.8). Uniqueness holds because the coefficient of I^1 has constant term -1 . $\square$

Corollary 7.5 (An alternating Narayana formula for the cores). *Write $i_{n,k} = [x^n][u^k]I(x, u)$ and let $v(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k+1}$ be the Narayana number of (5.3), taken to be zero outside $n \geq 1$, $0 \leq k < n$. Then, for $n \geq 2$ and $k \geq 1$,*

$$i_{n,k} = \sum_{d=0}^{k-1} (-1)^d \binom{k-1}{d} v(n-d-1, k-d-1). \quad (7.10)$$

Proof. From $J = xu(C - 1)$ and (5.3), $[x^m][u^\ell]J = v(m-1, \ell-1)$. Hence

$$I(x, u) = \sum_{m \geq 2} \sum_{\ell \geq 1} v(m-1, \ell-1) x^m u^\ell (1 + xu)^{-\ell},$$

and expanding $(1 + xu)^{-\ell} = \sum_{d \geq 0} (-1)^d \binom{\ell+d-1}{d} x^d u^d$, then setting $m = n - d$ and $\ell = k - d$, gives (7.10). $\square$

The sum alternates, but its value is a nonnegative integer because it counts cores. For example (7.10) gives $[x^4]I(x, u) = u + 2u^2$ and $[x^5]I(x, u) = u + 5u^2 + u^3$. The three cores of length four are indeed 4123, with one excedance, and 2413 and 3142, with two.

Corollary 7.6 (Components and the singleton). *Let $m(\pi)$ be the number of nontrivial direct-sum components of π and $s(\pi) \in \{0, 1\}$ its number of singleton components. Then*

$$\sum_{\pi} x^{|\pi|} u^{\text{exc}(\pi)} w^{m(\pi)} v^{s(\pi)} = \frac{1}{1 - w I(x, u)} + \frac{vx}{(1 - w I(x, u))^2}, \quad (7.11)$$

the sum being over the permutations counted by a_n . In particular the members with m nontrivial components and no singleton are counted by $I(x, u)^m$, and those with one singleton by $(m + 1)x I(x, u)^m$.

Proof. This is the decomposition behind (6.5), with the two additional markers carried along. No interaction term is needed: by theorem 2.3(iii) and theorem 4.1 no upper succession straddles a direct-sum boundary, so the components are chosen independently. Note that a singleton contributes the weight x and no excedance, which is why v appears multiplied by x alone. $\square$

8 Exact coefficients: recurrences and finite sums

8.1 A cancellation-free quadratic recurrence

The radical expression for I is equivalent to

$$\boxed{I = x^2 + (x + x^2)I + (1 + x)I^2, \quad I(0) = 0.} \quad (8.1)$$

Its unique solution starts at degree two. Write $i_n = [x^n]I$ and $h_n = [x^n]H$, and take negatively indexed coefficients to be zero. Then $i_0 = i_1 = 0$, $h_0 = 1$, and, for $n \geq 2$,

$$i_n = \mathbf{1}_{n=2} + i_{n-1} + i_{n-2} + \sum_{j=0}^n i_j i_{n-j} + \sum_{j=0}^{n-1} i_j i_{n-1-j}. \quad (8.2)$$

The apparent terms involving i_n vanish because $i_0 = i_1 = 0$. For $n \geq 1$,

$$h_n = \sum_{j=2}^n i_j h_{n-j}, \quad (8.3)$$

$$a_n = h_n + \sum_{j=0}^{n-1} h_j h_{n-1-j}, \quad a_0 = 1. \quad (8.4)$$

These formulas use only addition and multiplication of nonnegative integers. They compute the first N terms in $O(N^2)$ arithmetic operations and $O(N)$ stored integers; this is an arithmetic-operation bound, not a claim about bit complexity for growing integer sizes.

Table 1: The building-block sequences and the first terms of A289587.

n	i_n	h_n	$[x^n]B$	a_n
0	0	1	1	1
1	0	0	1	1
2	1	1	2	1
3	1	1	4	3
4	3	4	10	6
5	7	9	26	18
6	19	27	72	47
7	53	76	206	139
8	153	227	606	405
9	453	683	1820	1225
10	1367	2096	5558	3740

8.2 A linear-arithmetic coefficient algorithm

The recurrences (8.2)–(8.4) are convolutions. The radical form (1.3) supports a much cheaper signed alternative, in which the number of integer arithmetic operations is linear rather than quadratic. Write

$$T(x) = \sqrt{D(x)} = \sum_{n \geq 0} t_n x^n, \quad (d_0, d_1, d_2, d_3, d_4) = (1, -2, -5, -2, 1).$$

Proposition 8.1 (Holonomic recurrence for the radical). *With $t_n = 0$ for $n < 0$,*

$$t_0 = 1, \quad 2n t_n = \sum_{k=1}^{\min(4,n)} (3k - 2n) d_k t_{n-k} \quad (n \geq 1). \quad (8.5)$$

Every t_n is an integer, and every division indicated is exact.

Proof. Differentiating $T^2 = D$ gives $2DT' = D'T$. Comparing coefficients of x^{n-1} ,

$$2 \sum_{k=0}^4 d_k (n-k) t_{n-k} = \sum_{k=0}^4 k d_k t_{n-k}.$$

The $k = 0$ term contributes $2nt_n$ on the left and nothing on the right, because $d_0 = 1$; collecting the remaining terms gives (8.5). Integrality is not an accident of the recurrence: (6.3) rearranges to $T = 1 - x - x^2 - 2(1+x)I(x)$, and I has integer coefficients. $\square$

Theorem 8.2 (Linear-arithmetic coefficients). *Let $p_j = [x^j]P$, so $(p_0, \dots, p_4) = (3, 10, 11, 4, 1)$ and $p_j = 0$ for $j > 4$. With negative subscripts read as zero,*

$$a_n = \frac{p_{n+1} - 3t_{n+1} - 5t_n - t_{n-1}}{8} - 2a_{n-1} - a_{n-2} \quad (n \geq 0). \quad (8.6)$$

All divisions by 8 are exact.

Proof. Theorem 1.2 states $8x(1+x)^2A = P - QT$ with $Q = 3 + 5x + x^2$. Comparing coefficients of x^{n+1} gives $8(a_n + 2a_{n-1} + a_{n-2}) = p_{n+1} - 3t_{n+1} - 5t_n - t_{n-1}$, which is (8.6). Exactness of the division follows from the fact that the a_n count permutations. Compute $t_1 = -1$ from (8.5) before using (8.6) at $n = 0$, which then returns $a_0 = 1$. $\square$

There are at most four summands in (8.5) and a fixed number of operations in (8.6), so all terms through index N cost $O(N)$ integer arithmetic operations, against the $O(N^2)$ of (8.2)–(8.4). As before this is an arithmetic-operation bound and not a claim of $O(N)$ bit complexity. Section 9 shows that a_n has $\Theta(n)$ bits, so the complete output through index N occupies $\Theta(N^2)$ bits; a streaming implementation keeps only a fixed number of recent large integers and therefore needs $O(N)$ working bits beyond its output. The supplied programs keep whole arrays instead, for convenient comparison. The two routes are genuinely independent checks on one another: (8.2)–(8.4) use nonnegative integers only and no cancellation, while (8.5)–(8.6) are heavily signed.

8.3 A minimal polynomial for the target series

Squaring the radical formula and simplifying gives

$$4x(1+x)^2A^2 - P(x)A + (1+x)(2x^2 + 6x + 3) = 0. \quad (8.7)$$

The coefficient of A at $x = 0$ is -3 , so the branch $A(0) = 1$ is uniquely determined as a formal series. This provides another exact certificate of the formula. Since D has distinct roots and Q is not zero, $\sqrt{D}$ is not rational and neither is A ; the algebraic degree over $\mathbb{Q}(x)$ is exactly two. In particular, the counting sequence does not satisfy any eventual constant-coefficient linear recurrence.

8.4 A positive finite-sum formula

Let $\text{Cat}_j = (j+1)^{-1} \binom{2j}{j}$. Solving (8.1) in Catalan form yields

$$I(x) = \sum_{j \geq 0} \text{Cat}_j \frac{x^{2j+2}(1+x)^j}{(1-x-x^2)^{2j+1}}. \quad (8.8)$$

This follows either by substituting the Catalan equation or by factoring the discriminant as

$$D = (1-x-x^2)^2 \left(1 - \frac{4x^2(1+x)}{(1-x-x^2)^2} \right).$$

Expanding the two remaining factors gives, for $n \geq 2$,

$$i_n = \sum_{j=0}^{\lfloor (n-2)/2 \rfloor} \text{Cat}_j \sum_{\ell=0}^j \binom{j}{\ell} \sum_{\substack{a,b \geq 0 \\ a+2b=n-2j-2-\ell}} \frac{(2j+a+b)!}{(2j)! a! b!}. \quad (8.9)$$

Indeed, the coefficient of x^{a+2b} in $(1-x-x^2)^{-m}$ contributed by a copies of x and b copies of x^2 is $(m+a+b-1)!/((m-1)!a!b!)$.

The same rational factors can be expanded a second way, with one summation index instead of two. Since $(1-x-x^2)^{-1}$ is the Fibonacci generating function, this is a Fibonacci-power identity: for integers $h \geq 1$ and $m \geq 0$,

$$[x^m](1-x-x^2)^{-h} = \sum_{\ell=0}^{\lfloor m/2 \rfloor} \binom{h+m-\ell-1}{m-\ell} \binom{m-\ell}{\ell}. \quad (8.10)$$

To see this, expand first in powers of $x+x^2$, which contributes $\binom{h+j-1}{j}$ for the j th power, and then choose which ℓ of those j factors supply x^2 rather than x ; writing $j = m - \ell$ gives (8.10). Together with the finite expansion of $(1+x)^j$, identity (8.10) turns (8.8) into a nonalternating binomial formula for i_n with one fewer summation than (8.9). Both are recorded because they are independent evaluations of the same series.

For a completely recurrence-free formula for a_n , let $C_{\geq 2}(m, k)$ be the ordered k -tuples of integers at least two whose sum is m . For $k = 0$ it consists of the empty tuple if $m = 0$ and is empty otherwise. Then

$$\begin{aligned} a_n = & \sum_{k=0}^{\lfloor n/2 \rfloor} \sum_{(m_1, \dots, m_k) \in C_{\geq 2}(n, k)} \prod_{j=1}^k i_{m_j} \\ & + \sum_{k=0}^{\lfloor (n-1)/2 \rfloor} (k+1) \sum_{(m_1, \dots, m_k) \in C_{\geq 2}(n-1, k)} \prod_{j=1}^k i_{m_j}. \end{aligned} \quad (8.11)$$

The second outer sum is empty at $n = 0$. Together with (8.9), this is a finite formula involving only binomial coefficients, factorials, and nonnegative sums and products. It follows directly from

$$A = \sum_{k \geq 0} I^k + x \sum_{k \geq 0} (k+1) I^k.$$

The factor $k+1$ is also the number of places to insert one singleton among k nontrivial blocks.

8.5 An independent alternating coefficient formula

The Narayana substitution provides a different computational route: for $n \geq 1$,

$$[x^n]B(x) = 1 + \sum_{m=2}^n \sum_{k=1}^{m-1} (-1)^{n-m} \frac{1}{m} \binom{m}{k} \binom{m}{k+1} \binom{k+n-m-1}{n-m}. \quad (8.12)$$

Here the initial 1 accounts for zero excedances. This follows by expanding $(1+x)^{-k}$ in $B(x) = C(x, (1+x)^{-1})$. Unlike (8.2)–(8.4), it has cancellation, but precisely for that reason it is useful as an independent exact-arithmetic check.

9 Asymptotic enumeration

The radical formula also determines the exponential growth and its polynomial correction. Two complementary ways of locating the zeros of D are worth recording. The first is an explicit real factorization,

$$D(x) = (x^2 - (1 + 2\sqrt{2})x + 1)(x^2 + (2\sqrt{2} - 1)x + 1). \quad (9.1)$$

The second uses the fact that D is reciprocal: for $x \neq 0$,

$$x^{-2}D(x) = (x + x^{-1})^2 - 2(x + x^{-1}) - 7, \quad (9.2)$$

so the zeros satisfy $x + x^{-1} = 1 \pm 2\sqrt{2}$. The plus sign gives a positive reciprocal pair; the minus sign gives $|1 - 2\sqrt{2}| < 2$ and hence a conjugate pair on the unit circle. Either route shows that the smallest-modulus zero is the positive simple number

$$\rho = \frac{1 + 2\sqrt{2} - \sqrt{5 + 4\sqrt{2}}}{2} = 0.281971680061194853146616 \dots \quad (9.3)$$

The rational denominator in (1.3) has no nonremovable zero inside the unit disk. Its numerator has a nonzero square-root coefficient at ρ , so ρ is the unique dominant singularity.

Theorem 9.1 (Precise leading asymptotic). *Let*

$$\gamma = \rho^{-1} = \frac{1 + 2\sqrt{2} + \sqrt{5 + 4\sqrt{2}}}{2} = 3.546455444684995244456761 \dots, \quad (9.4)$$

$$\delta = \sqrt{-\rho D'(\rho)}, \quad K = \frac{Q(\rho) \delta}{16\sqrt{\pi} \rho (1 + \rho)^2} = 0.413931068036970644808821 \dots, \quad (9.5)$$

where $Q(\rho) = \rho^2 + 5\rho + 3 > 0$, so the square-root term does not cancel. Then

$$a_n \sim K \gamma^n n^{-3/2}. \quad (9.6)$$

Proof. Put $z = 1 - x/\rho$. The root is simple and $D'(\rho) < 0$, so

$$\sqrt{D(x)} = \delta z^{1/2} (1 + O(z)).$$

Write $A = R - S\sqrt{D}$, where

$$R(x) = \frac{P(x)}{8x(1+x)^2}, \quad S(x) = \frac{Q(x)}{8x(1+x)^2}.$$

The leading nonanalytic term is $-S(\rho)\delta z^{1/2}$. The binomial-coefficient asymptotic

$$[x^n](1 - x/\rho)^{1/2} = -\frac{\rho^{-n}}{2\sqrt{\pi} n^{3/2}}(1 + O(n^{-1}))$$

follows from the gamma-function expression for $(-1)^n \binom{1/2}{n}$ and Stirling's formula. A slit neighborhood extends to a disk of radius strictly larger than ρ without meeting another singularity. The convergent local square-root expansion therefore transfers termwise; the next nonanalytic term is of order $z^{3/2}$ and contributes $O(\rho^{-n}n^{-5/2})$. Equivalently, this is the usual square-root singularity transfer [4], obtained by deforming a Cauchy coefficient contour to the slit at ρ . Thus the leading constant is $S(\rho)\delta/(2\sqrt{\pi})$, namely (9.5). $\square$

The first relative correction is also explicit. Define

$$c_1 = \frac{3}{8} - \frac{3}{2} \left(\frac{\rho^2 D''(\rho)}{4\delta^2} - \frac{\rho S'(\rho)}{S(\rho)} \right) = -1.008253037811239547502410 \dots \quad (9.7)$$

Since $\delta^2 = -\rho D'(\rho)$, this is the same number as

$$c_1 = \frac{3}{8} + \frac{3\rho S'(\rho)}{2S(\rho)} + \frac{3\rho D''(\rho)}{8D'(\rho)}, \quad (9.8)$$

which is the form the programs evaluate. Expanding one term further gives

$$a_n = K\gamma^n n^{-3/2} \left(1 + \frac{c_1}{n} + O(n^{-2}) \right). \quad (9.9)$$

For completeness, the coefficient of z in $S(\rho(1-z))\sqrt{D(\rho(1-z))}/(S(\rho)\delta\sqrt{z})$ is the parenthesized expression in (9.7). The correction $3/8$ comes from the coefficient asymptotic for $z^{1/2}$, while the factor $-3/2$ is $\Gamma(-1/2)/\Gamma(-3/2)$.

Table 2: Exact coefficients against the one- and two-term asymptotics. The middle column should converge to c_1 , the last to 1.

n	$a_n/(K\gamma^n n^{-3/2})$	$n(a_n/(K\gamma^n n^{-3/2}) - 1)$	$a_n/(K\gamma^n n^{-3/2}(1 + c_1/n))$
10	0.907817390152669...	-0.9218260984733...	1.009611807327...
20	0.951983446713366...	-0.9603310657327...	1.002523305106...
25	0.961218942324211...	-0.9695264418947...	1.001614163236...
50	0.980230192459601...	-0.9884903770200...	1.000403387550...
100	0.990017308750539...	-0.9982691249461...	1.000100856012...
200	0.994983824600422...	-1.0032350799160...	1.000025216915...
250	0.995983062047636...	-1.0042344880910...	1.000016139289...
500	0.997987520864276...	-1.0062395678620...	1.000004035077...
1000	0.998992754748870...	-1.0072452511300...	1.000001008804...

The last column shrinks towards 1 at the rate n^{-2} predicted by (9.9), which is what a merely fitted correction term would not be expected to do.

Two consequences. First, $\log_2 a_n = n \log_2 \gamma + O(\log n)$, so a_n occupies $\Theta(n)$ bits; this is the size estimate used in section 8.2. Second, since $|\text{Av}_n(321)| = \text{Cat}_n \sim 4^n/(\sqrt{\pi} n^{3/2})$, the proportion of 321-avoiders meeting the mesh restriction is

$$\frac{a_n}{\text{Cat}_n} \sim K\sqrt{\pi} \left(\frac{\gamma}{4} \right)^n. \quad (9.10)$$

The extra geometric condition is therefore exponentially selective inside the Catalan class, while the characteristic exponent $n^{-3/2}$ survives.

Corollary 9.2 (A limiting fixed-point probability). *Choose an A289587 permutation uniformly at random from those of length n . The probability that it has exactly one fixed point, equivalently exactly one singleton direct-sum component, tends to*

$$\frac{2\rho H(\rho)}{1 + 2\rho H(\rho)} = \frac{2\rho}{1 - I(\rho) + 2\rho} = 0.428885255667078255465240\dots, \quad (9.11)$$

where

$$H(\rho) = \frac{1 + 3\rho + \rho^2}{4\rho(1 + \rho)}, \quad I(\rho) = \frac{1 - \rho - \rho^2}{2(1 + \rho)} = 0.249038376398374331486125\dots \quad (9.12)$$

Proof. Write $H = h_0 - h_1\sqrt{1 - x/\rho} + O(1 - x/\rho)$, where $h_1 > 0$. The corresponding square-root coefficients in $H + xH^2$ and xH^2 are $-h_1(1 + 2\rho h_0)$ and $-2\rho h_0 h_1$. Their coefficient ratio gives the first expression in (9.11); singularity transfer applies to both summands separately. The second expression follows from $H = 1/(1 - I)$, and $I(\rho)$ is obtained by putting $D(\rho) = 0$ in (6.3). $\square$

10 Reproducibility and proof audit

The archive contains *two* independent standard-library Python verifiers, living in code/:

- `verify.py`, with the optional SymPy/mpmath companion `symbolic_checks.py`;
- `verify_records.py`, with the optional companion `symbolic_check_records.py`.

The two verifiers were written separately and share no code. They are independent implementations of the same test design: each generates Av(321) by inserting the new maximum, and each then has its own mesh predicate, its own contraction routine and its own coefficient algorithm, the first organized around excedances and the second around left-to-right maxima. What is independent is the implementation, not the generation strategy. That duplication is deliberate and is retained on purpose; a third short script, `code/cross_check.py`, compares their four coefficient algorithms against each other. No program downloads data or uses numerical guessing to establish the formula.

Independent enumeration. To generate all 321-avoiders, insert a new maximum into every position whose following suffix is increasing. A newly created 321 would have to start at the new maximum, so this criterion is necessary and sufficient. Every resulting permutation has a unique parent obtained by deleting its maximum. Catalan totals are checked at each length, and the second verifier additionally compares its generated sets against a filter of all $n!$ permutations through $n = 8$. For mesh membership, both programs test the nine-rectangle definition directly, rather than replacing it by theorem 3.3. This is important because the structural theorem is exactly the result that the independent tests should be capable of falsifying.

The first verifier carries the literal nine-rectangle test three lengths further, to $n = 12$ for both patterns and to $n = 13$ structurally. Table 3 additionally exhibits the two mesh columns side by side, which is the finite counterpart of theorem 1.1; the second verifier also checks that inversion matches the two occurrence sets pointwise through $n = 8$.

All checks passed. At length 12, the literal test examines all 208012 Catalan permutations and counts 36357 accepted ones; at length 13, the structural test examines all 742900 Catalan permutations and counts 115049.

Table 3: Direct geometric enumeration by the second verifier. These columns are counted from the nine-rectangle definition, not expanded from the conjectured formula. The last column counts the nontrivial sum-indecomposable bond-free cores, i_n .

n	$ \text{Av}_n(321) $	avoid $(12, R_{174})$	avoid $(12, R_{234})$	i_n
0	1	1	1	0
1	1	1	1	0
2	2	1	1	1
3	5	3	3	1
4	14	6	6	3
5	42	18	18	7
6	132	47	47	19
7	429	139	139	53
8	1430	405	405	153
9	4862	1225	1225	453
10	16796	3740	3740	1367
11	58786	11602	11602	4191

The second verifier examines all 82500 Catalan permutations of lengths 0 through 11. The first symbolic script verifies seven algebraic residuals identically equal to zero, including (6.5), (8.7), and (9.1); the second certifies (6.7) and (6.8) as exact polynomial divisions with the quotients $4(1+x)$ and $2(Q(x)r - 3x^2 - 7x - 3)$ displayed in section 6, and evaluates the constants of section 9 at 90 digits. Both sets of constants agree to every digit printed here. The coefficient files extend the sequence through $n = 200$ and through $n = 1000$; both are explicitly identified as computed from this proof and neither is an official OEIS file. For orientation the extension begins

n	a_n	n	a_n
17	12380078	22	4778632088
18	40423803	23	15884348886
19	132564051	24	52941973417
20	436417933	25	176889898401
21	1441809621	26	592374142673

These tests guard against errors in the shading convention, in the bijections, in the refinements and in the algebra. They are not a proof of an identity for all n .

What constitutes the proof. Finite tests are corroboration, not a substitute for the argument. The proof consists of four logically separate steps: the exact mesh occurrence classification; reversible upper-run contraction; the Narayana functional equation with an excedance marker; and the fixed-point/direct-sum decomposition. All later identities are algebraic consequences of those steps. There is no extrapolation from a guessed recurrence, no unverified infinite-state truncation, and no dependence on checking only the currently displayed OEIS terms.

Rebuilding. From the archive directory, run

```
python3 code/verify.py
```

Table 4: Executed checks included in the reproducibility logs. The first group is `verify.py`, the second `verify_records.py`, the third `cross_check.py`.

Check	Range
Literal mesh test, both R_{174} and R_{234} , against the structural characterization and coefficient recurrence	$0 \leq n \leq 12$
Structural counts and every fixed-point refinement against the coefficient recurrences	$0 \leq n \leq 13$
Core contraction/inflation round trips and exact occurrence identity (3.2)	$0 \leq n \leq 10$
Radical expansion versus the positive recurrences	$0 \leq n \leq 200$
Narayana-substitution finite sum (8.12)	$0 \leq n \leq 60$
Positive finite sum (8.9)	$0 \leq n \leq 45$
Published OEIS initial values	$0 \leq n \leq 16$
Generator against a filter of all $n!$ permutations	$0 \leq n \leq 8$
Literal mesh test, both patterns, against the structural characterization	$0 \leq n \leq 11$
Pointwise inversion of mesh occurrences, theorem 1.1	$0 \leq n \leq 8$
Record-to-Dyck bijection round trips and the Narayana distribution of left-to-right maxima	$0 \leq n \leq 11$
Refined core formula (7.10) and the full refinement (7.11) against joint counts	$0 \leq n \leq 11$
Contraction/inflation round trips on indecomposables	23713 round trips
Independent inflations of small cores	102 tests
All four coefficient algorithms against each other, including (8.5)–(8.6) against (8.2)–(8.4)	$0 \leq n \leq 1000$

```
python3 code/verify_records.py --max-n 1000 --exhaustive 11
python3 code/cross_check.py
python3 code/symbolic_checks.py          # optional; needs sympy, mpmath
python3 code/symbolic_check_records.py   # optional; needs sympy, mpmath
pdflatex -interaction=nonstopmode article.tex
pdflatex -interaction=nonstopmode article.tex
```

Do not use Python's `-O` flag, which disables the assertions that are the checks. The verifiers write coefficient tables and machine-readable reports to `data/` under distinct names; the optional scripts write exact symbolic check results and high-precision numerical approximations. Raising the exhaustive limit entails Catalan growth and is unrelated to the linear cost of generating coefficients. This note provides a conventional mathematical proof; it is not a proof-assistant formalization, and its numerical approximations are high-precision evaluations of exact expressions, not interval-arithmetic certificates.

11 Conclusion

The generating-function conjecture for A289587 is true. The proof rests on an exact classification of mesh occurrences as fixed-point pairs or pairs in one upper run. Reversible run contraction gives $B(x, u) = C(x, u/(1 + xu))$, and the direct-sum decomposition gives $H = B/(1 + xB)$ and $A = H + xH^2$. Thus the conjectured quartic radical comes from the Narayana equation after a canonical change of variables, not from extrapolating a finite coefficient list. The same structure yields refined occurrence enumeration, positive

recurrences and finite formulas, a coefficient algorithm of linear arithmetic cost, and a precise asymptotic expansion with its $1/n$ correction.

References

- [1] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A289587, <https://oeis.org/A289587>. Sequence initiated by N. J. A. Sloane; generating-function conjecture attributed in the entry to Thomas Scheuerle, December 23, 2025. Entry consulted September 20, 2026.
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- [3] Petter Brändén and Anders Claesson, *Mesh patterns and the expansion of permutation statistics as sums of permutation patterns*, Electronic Journal of Combinatorics **18**(2) (2011), Paper P5, <https://doi.org/10.37236/2001>. Preprint: <https://arxiv.org/abs/1102.4226>. The foundational source for the general mesh-pattern framework; it is not a source for the specific argument given here.
- [4] Philippe Flajolet and Robert Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, <https://ac.cs.princeton.edu/home/>. Background for formal Lagrange inversion and for the transfer of square-root singular expansions. The functional equations and the local expansion used above are derived here explicitly, and the dominant-singularity hypotheses are checked.
- [5] Astrid Reifegerste, *The excedances and descents of bi-increasing permutations*, preprint, 2002, arXiv:math/0212247, <https://arxiv.org/abs/math/0212247>. In particular, the bi-increasing characterization, the fixed-point separation property, and Corollary 3.7 on the Narayana distribution provide background for the self-contained preliminaries above.